

GEOS-MPM Verification Suite Report

Generated by verification/runVerificationSuite

May 18, 2026

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1 Running and rerunning

Cases are staged under `testRunDirectory/verification/<case>`. The suite records failures and continues unless `--stop-on-error` is supplied. Use `--force` for noninteractive reruns.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/verification
./runVerificationSuite all --python /usr/tce/bin/python3 --force
./runVerificationSuite report --python /usr/tce/bin/python3
```

2 Timing and dispatch summary

Case	Status	Family	GEOS job	GEOS state	GEOS wall	Issue
anneal__annealCompression	failed	Annealing	6059617	COMPLETED	00:00:04	failure_marker
anneal__annealTension	failed	Annealing	6059619	COMPLETED	00:00:04	failure_marker
cohesiveZones__cz_gridAligned	failed	Cohesive zones	6059621	COMPLETED	00:00:07	failure_marker
geomechanics__geoModel.FTable	failed	Geomechanics constitutive verification	6059623	COMPLETED	00:00:17	failure_marker
implicitFluid__fluidPressure	failed	Implicit fluid/pressure	6059625	COMPLETED	00:00:04	failure_marker

3 Failures, expected failures, and diagnostics

3.1 anneal__annealCompression

Status: failed Issue: failure_marker

```
--- anneal__annealCompression_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in Anneal (mpm_annealCompression.xml, 1.66): XML Node at '/Problem/Solvers/SolidMechanics_MPM/MPMEvents/Anneal' contains unused attribute 'source'.
[2026-05-18T20:15:32.838] error: *** STEP 6059617.0 ON dane86 CANCELLED AT 2026-05-18T20:15:32 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```

3.2 anneal__annealTension

Status: failed Issue: failure_marker

```
--- anneal__annealTension_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in Anneal (mpm_annealTension.xml, 1.66): XML Node at '/Problem/Solvers/SolidMechanics_MPM/MPMEvents/Anneal' contains unused attribute 'source'.
[2026-05-18T20:16:43.024] error: *** STEP 6059621.0 ON dane86 CANCELLED AT 2026-05-18T20:16:43 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```

3.3 cohesiveZones__cz_gridAligned

Status: failed Issue: failure_marker

```
--- cohesiveZones__cz_gridAligned_geos_slurm_check.log ---
failure_marker=XML parsing error
***** XML parsing error at node named CoupledCohesiveZone, attribute defaultMaxNormalStress (mpm_cz_gridAligned.xml, 1.95).
***** EXCEPTION
***** Error cause: !m_successfulReadFromInput
***** Rank 0: XML Node CoupledCohesiveZone (mpm_cz_gridAligned.xml, 1.95) with name=cz1 is missing required attribute 'defaultMaxNormalStress'.
[2026-05-18T20:16:43.024] error: *** STEP 6059621.0 ON dane86 CANCELLED AT 2026-05-18T20:16:43 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```

3.4 geomechanics__geoModel_FTable

Status: failed Issue: failure_marker

```
--- geomechanics__geoModel_FTable_plotReactions_geomechanicsTXC.log ---
Traceback (most recent call last):
```

3.5 implicitFluid__fluidPressure

Status: failed Issue: failure_marker

```
--- implicitFluid__fluidPressure_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in mpsolve (mpm_fluidPressure.xml, 1.45): XML Node at '/Problem/Solvers/SolidMechanics_MPM' contains unused attribute 'implicitFluidPressure'.
[2026-05-18T20:15:32.840] error: *** STEP 6059625.0 ON dane86 CANCELLED AT 2026-05-18T20:15:32 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```

4 Case studies

4.1 anneal / annealCompression

This verification problem exercises annealing using the mesh, particle seeding, material model, boundary conditions, and history outputs defined in `pfw_input_annealCompression.py`. The plots below include initial and final state renderings plus the reaction and box-average histories needed for analytical comparison.

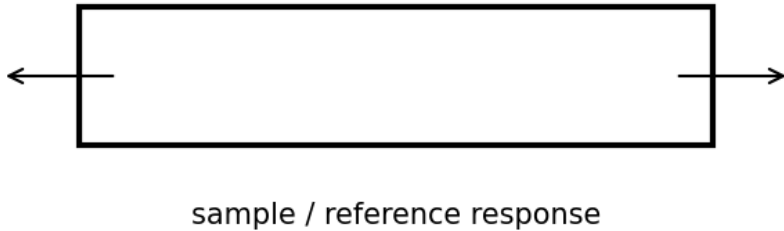


Figure 1: Geometry-aware schematic for anneal_annealCompression.

Code features exercised.

- FLIP/PIC/XPIC transfer
- reaction-history output
- box-average history
- ElasticIsotropic material
- temperature or annealing event

Verification plotting plan.

Script/source	Data source	Variables	Processing
generic-history-plots	reactionHistory.csv when present; boxAverageHistory.csv when present	reaction/force components; stress/s- train/pressure/plastic/damage/ve- locity/energy/momentum columns	pressure = components nent magni strain/mom triplets are

missing initial/final/history plots

```

--- anneal__annealCompression_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in Anneal (mpm_annealCompression.xml, 1.66): XML Node at '/Problem/Solvers/SolidMechanics_MPM/MPMEvents/Anneal' contains unused attribute 'source'.
[2026-05-18T20:15:32.838] error: *** STEP 6059617.0 ON dane86 CANCELLED AT 2026-05-18T20:15:32 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed

```

4.2 anneal / annealTension

This verification problem exercises annealing using the mesh, particle seeding, material model, boundary conditions, and history outputs defined in pfw_input_annealTension.py. The plots below include initial and final state renderings plus the reaction and box-average histories needed for analytical comparison.

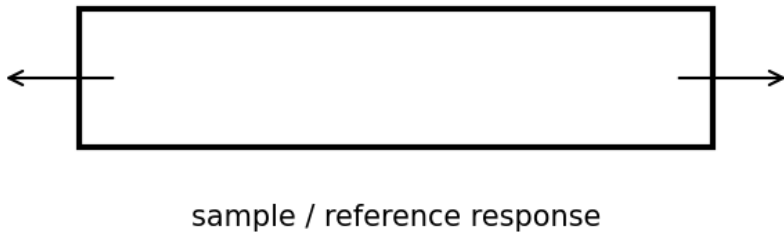


Figure 2: Geometry-aware schematic for anneal_annealTension.

Code features exercised.

- FLIP/PIC/XPIC transfer
- reaction-history output
- box-average history
- ElasticIsotropic material
- temperature or annealing event

Verification plotting plan.

Script/source	Data source	Variables	Processing
generic-history-plots	reactionHistory.csv when present; boxAverageHistory.csv when present	reaction/force components; stress/s- train/pressure/plastic/damage/ve- locity/energy/momentum columns	pressure = components nent magni- strain/mom- triplets are

missing initial/final/history plots

```
--- anneal__annealTension_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in Anneal (mpm_annealTension.xml, 1.66): XML Node at '/Problem/Solvers/SolidMechanics_MPM/MPMEvents/Anneal' contains unused attribute 'source'.
[2026-05-18T20:15:32.838] error: *** STEP 6059619.0 ON dane86 CANCELLED AT 2026-05-18T20:15:32 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```

4.3 cohesiveZones / cz gridAligned

This verification problem exercises cohesive zones using the mesh, particle seeding, material model, boundary conditions, and history outputs defined in pfw_input_cz_gridAligned.py. The plots below include initial and final state renderings plus the reaction and box-average histories needed for analytical comparison.

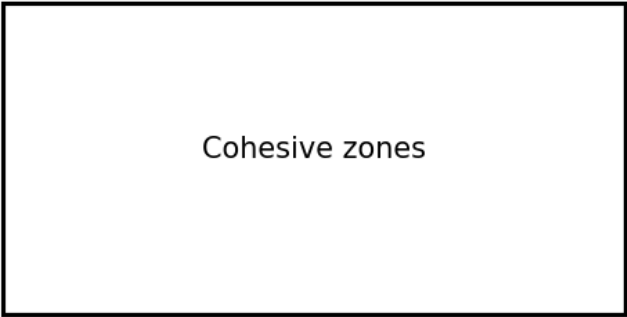


Figure 3: Geometry-aware schematic for cohesiveZones__cz_gridAligned.

Code features exercised.

- FMPM update
- FLIP/PIC/XPIC transfer
- periodic boundaries
- reaction-history output
- box-average history
- cohesive-zone fields
- ElasticIsotropic material
- contact/multi-field interaction

Verification plotting plan.

Script/source	Data source	Variables	Processing
generic-history-plots	reactionHistory.csv when present; boxAverageHistory.csv when present	reaction/force components; stress/s- train/pressure/plastic/damage/ve- locity/energy/momentum columns	pressure = components nent magni strain/mom triplets are

missing initial/final/history plots

```
--- cohesiveZones__cz_gridAligned_geos_slurm_check.log ---
failure_marker=XML parsing error
***** XML parsing error at node named CoupledCohesiveZone, attribute defaultMaxNormalStress (mpm_cz_gridAligned.xml, 1.95).
***** EXCEPTION
***** Error cause: !m_successfulReadFromInput
***** Rank 0: XML Node CoupledCohesiveZone (mpm_cz_gridAligned.xml, 1.95) with name=cz1 is missing required attribute 'defaultMaxNormalStress'.
[2026-05-18T20:16:43.024] error: *** STEP 6059621.0 ON dane1511 CANCELLED AT 2026-05-18T20:16:43 DUE to SIGNAL Killed ***
slurm: error: dane1511: task 0: Killed
```

4.4 geomechanics / geoModel FTable

This verification problem exercises geomechanics constitutive verification using the mesh, particle seeding, material model, boundary conditions, and history outputs defined in `pfw_input_geoModel_FTable.py`. The plots below include initial and final state renderings plus the reaction and box-average histories needed for analytical comparison.

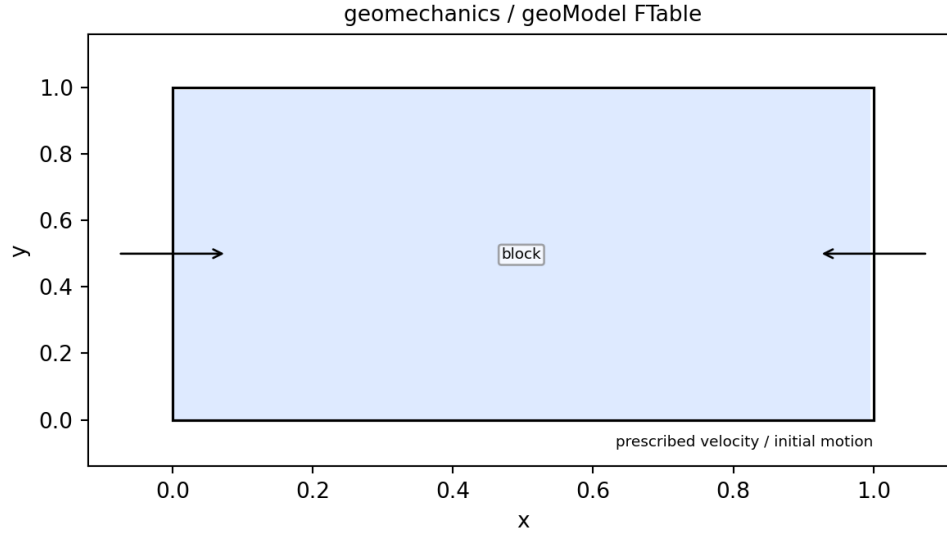


Figure 4: Geometry-aware schematic for `geomechanics__geoModel_FTable`.

Code features exercised.

- FLIP/PIC/XPIC transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model

Script/source	Data source	Variables	Processing
plotReactions_geomechanicsBucklingHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; pressure; energy	derive stress force divide by solid volume; compute or vector/t pressure or stress compute
plotReactions_geomechanicsDeformationHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; pressure; energy	derive stress force divide by solid volume; compute or vector/t pressure or stress compute
plotReactions_geomechanicsHydrolysisHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; pressure; energy	derive stress force divide by solid volume; compute or vector/t pressure or stress compute
plotReactions_geomechanicsTXCavityHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; plastic; pressure; energy	derive stress force divide by solid volume; compute or vector/t pressure or stress compute
plotReactions_geomechanicsTensionHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; plastic; pressure; energy	derive stress force divide by solid volume; compute or vector/t pressure or stress compute
plotReactions_geomechanicsUnconfinedBucklingHistory.csv;	boxAverageHistory.csv	reaction; stress; strain; damage; energy	derive stress force divide by solid volume; compute or vector/t

Verification plotting plan.

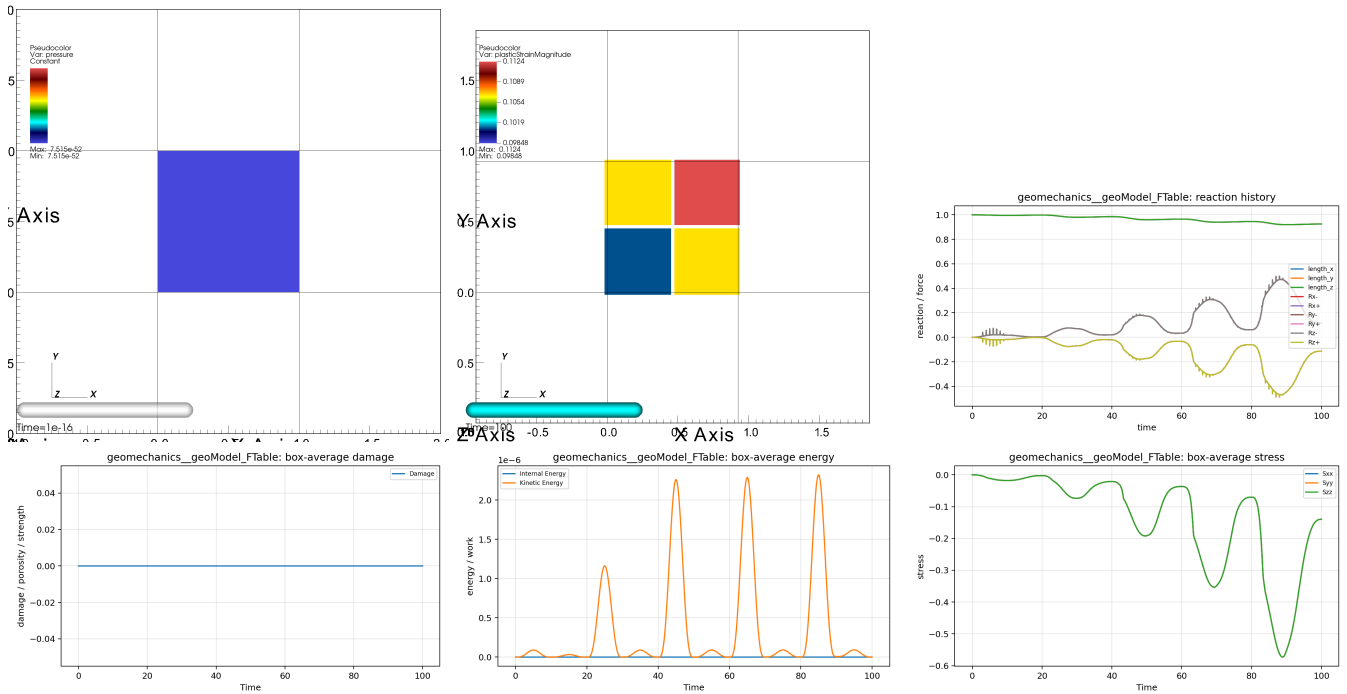


Figure 5: Initial/final state and verification history plots for geomechanics_geoModel_FTable.

```
--- geomechanics_geoModel_FTable_plotReactions_geomechanicsTXC.log ---
Traceback (most recent call last):
```

4.5 implicitFluid / fluidPressure

This verification problem exercises implicit fluid/pressure using the mesh, particle seeding, material model, boundary conditions, and history outputs defined in pfw_input_fluidPressure.py. The plots below include initial and final state renderings plus the reaction and box-average histories needed for analytical comparison.

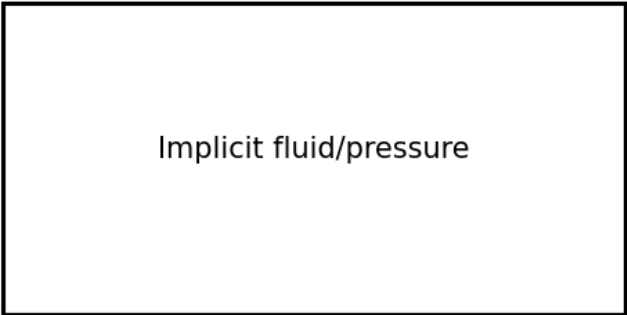


Figure 6: Geometry-aware schematic for implicitFluid__fluidPressure.

Code features exercised.

- FMPM update
- FLIP/PIC/XPIC transfer
- periodic boundaries
- reaction-history output
- box-average history
- ElasticIsotropic material
- contact/multi-field interaction

Verification plotting plan.

Script/source	Data source	Variables	Processing
generic-history-plots	reactionHistory.csv when present; boxAverageHistory.csv when present	reaction/force components; stress/s- train/pressure/plastic/damage/ve- locity/energy/momentum columns	pressure = components nent magni- strain/mom triplets are

missing initial/final/history plots

```
--- implicitFluid__fluidPressure_geos_slurm_check.log ---
failure_marker=EXCEPTION
***** EXCEPTION
***** Error cause: processedAttributes.count( attributeName ) == 0
***** Rank 0: Error in mpsolve (mpm_fluidPressure.xml, 1.45): XML Node at '/Problem/Solvers/SolidMechanics_MPM' contains unused attribute 'implicitFluidPressure'.
[2026-05-18T20:15:32.840] error: *** STEP 6059625.0 ON dane86 CANCELLED AT 2026-05-18T20:15:32 DUE to SIGNAL Killed ***
srun: error: dane86: task 0: Killed
```